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ORIGINAL ARTICLE

Digital health technology in oncology: results from a survey by the ESMO Young Oncologists Committee and medical student alumni

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Background and purpose: Digital health technologies are transforming oncology care, yet adoption remains inconsistent globally. This study evaluates current utilization patterns, perceived effectiveness, and barriers among oncology professionals worldwide.

Methods: We conducted a global cross-sectional survey (27 questions) analyzing workplace characteristics, technology usage patterns, effectiveness, and barriers to adoption among oncology professionals to better understand the real-world adoption of digital health technologies. Using a questionnaire developed by the ESMO Young Oncologists Committee in collaboration with the medical student alumni, we assessed device availability, technology usage, effectiveness ratings, training experiences, and implementation barriers. Responses were analyzed using descriptive statistical methods to identify usage patterns, variations and barriers to adoption of digital health technologies.

Results: A total of 423 oncology professionals completed the questionnaire. The survey revealed widespread access to computerized workstations (89.1%) but notable geographic disparities (Africa 67.7% versus Europe 93.8%, $P < 0.001$). Email remained the most widely used communication tool (51.8%), while video telehealth was rated as the most effective (62.9% scoring 4-5 on a 5-point Likert scale). Despite 46.8% of oncology professionals reporting using telemedicine systems, only 28.2% received formal training. Respondents reported an increase of adoption of digital health solutions in 93.3% of the cases. Older clinicians (>40 years) showed 12% higher video telehealth usage than younger colleagues (35% versus 23%, $P = 0.014$). Major barriers included system integration challenges (67.4%) and infrastructure limitations (57.3%), while 84.1% expressed a need for additional training. Effectiveness varied significantly by tool, from 80.5% for e-prescriptions to 45.2% for telemonitoring. Regional patterns emerged, with high-bandwidth tools predominant in well-resourced areas (Oceania 85.7% video telehealth) versus text-based systems where connectivity is more limited (Africa 63% chat-based).

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Discussion: Findings from the present survey highlight persistent disparities in digital health adoption, emphasizing the need for infrastructure investment, interoperable systems, and tailored training to ensure equitable implementation and maximize the potential of technology in oncology care globally.

Key words: digital health, oncology, technology

INTRODUCTION

The rapid evolution of digital health (DH) technologies is reshaping the landscape of oncology care, providing new opportunities for patient management, data analysis, and communication.[1] DH encompasses a range of tools, including telemedicine, electronic health records (EHRs), mobile health applications, wearable devices, and artificial intelligence (AI)-powered diagnostics.[2] These technologies have demonstrated significant potential to improve outcomes in cancer care by facilitating timely interventions, enhancing patient engagement, and optimizing resource utilization.[3]

Several studies have underscored the benefits of DH integration in oncology. For instance, telemedicine showed to improve access to specialized oncology care for patients in rural or underserved areas, reducing travel burdens and associated costs.[4,5] Similarly, AI-powered tools are increasingly being used to assist in diagnostic accuracy, treatment planning, and monitoring adverse events, thereby enhancing the accurate use of personalized medicine.[6,7] Furthermore, mobile health applications have proven efficacy in promoting treatment adherence and symptom tracking, enabling real-time communication between patients and health care providers.[8-11]

Despite these advancements, the adoption of DH solutions in oncology remains inconsistent across countries, regions, and health care settings. Barriers, including insufficient infrastructure, a lack of interoperability between systems, and limited digital literacy among health care providers and patients, hinder the widespread implementation of DH solutions.[12,13] Additionally, concerns related to data security, privacy, and the usability of digital platforms further complicate the integration process.[14,15] Notably, vulnerable populations, including the elderly and those from lower socioeconomic backgrounds, often face greater challenges in accessing and utilizing these technologies, exacerbating existing health disparities.[16,17]

The present study aimed to provide a comprehensive analysis of DH adoption in oncology, drawing on data from a global survey of oncology professionals. By identifying key trends, perceived barriers, and educational needs, we seek to inform strategies for promoting equitable and effective use of DH solutions in cancer care.

METHODS

This study employed a cross-sectional survey design targeting oncology health care professionals globally. The 'Oncology Healthcare Professional Digital Health Survey' was developed by alumni from the 2022 European Society for Medical Oncology (ESMO) Course in Medical Oncology for Medical Students, in collaboration with the ESMO Young Oncologists Committee, to evaluate the adoption and perceptions of DH technologies among oncology professionals. The survey link was distributed to 8632 eligible ESMO members. At the time of dissemination, 935 members had opted in to receive surveys. Owing to the survey's anonymous design, it was not possible to determine how many members actually received or opened the invitation. Accordingly, the response rate may be expressed as 4.9% of all contacted members or 45.2% of those who had opted in. The survey was available online for 60 days, from 26 March to 24 May 2024.

Survey design

The survey consisted of 27 questions distributed across multiple domains, including participant demographics, workplace characteristics, technology usage patterns, effectiveness, and barriers to adoption. Questions employed a mix of multiple-choice, Likert-scale, and open-ended formats. Key areas of focus included familiarity with DH tools (e.g. telemedicine, EHRs, mobile applications, AI-powered systems), perceived effectiveness and frequency of use of these technologies, barriers to adoption, such as integration challenges and infrastructure limitations, and educational needs and confidence levels in using DH solutions. The full survey is included as Supplementary Material, available at https://doi.org/10.1016/j.esmorw.2026.100697.

Distribution

The survey was disseminated online via ESMO channels, including email newsletters, social media platforms, and professional networks, to maximize reach among oncology health care professionals. To be eligible to complete the survey, respondents had to be ESMO members and an oncology health care provider. The survey was responded voluntarily by eligible subjects and the ones that answered the survey partially received an email asking them to complete the remaining questions. Responses were collected over 60 days.

Data analysis

Descriptive statistics summarized participant characteristics and technology usage patterns. Percentages were calculated according to the total of answers to each question; therefore, the denominator varies between questions. Inferential analyses, including chi-square tests and analysis of variance, were conducted to assess associations between demographic variables and DH adoption. Open-ended responses were thematically analyzed to identify recurring themes related to barriers and recommendations

Table 1. Participant demographics and professional characteristics (N = 423)

Characteristic	n (%)
Continent	
Europe	192 (45.4)
Asia	114 (27.0)
Americas	72 (17.0)
Africa	31 (7.3)
Oceania	14 (3.3)
Specialty	
Medical oncology	292 (69.0)
Clinical oncology	74 (17.5)
Radiation oncology	31 (7.3)
Surgical oncology	19 (4.5)
Pathologist	5 (1.2)
Other health care professional	2 (0.5)
Age group	
21-30 years	54 (12.8)
31-40 years	223 (52.7)
41-50 years	82 (19.4)
51-60 years	36 (8.5)
>60 years	11 (2.6)
Gender (male)	206 (48.8)
Training status (trainee)	116 (27.4)
Site of work	
Hospitals—all types	177 (41.8)
University/education institution	117 (27.7)
Comprehensive cancer centre	86 (20.3)
Private practice	30 (7.1)
Cancer research organization	8 (1.9)
Other	5 (1.2)

for improving DH integration. A $P < 0.05$ was considered statistically significant. Statistical analysis was carried out using Stata 14 software (StataCorp, College Station, TX).

RESULTS

Population characteristics

From an initial pool of 469 collected surveys, 423 responses contained sufficient data suitable for the analysis. As shown in Table 1, the respondent cohort comprised predominantly of medical oncologists (69.0%, $n = 292$), followed by clinical oncologists (17.5%, $n = 74$), radiation oncologists (7.3%, $n = 31$), surgical oncologists (4.5%, $n = 19$), pathologists (1.2%, $n = 5$), and other health care professionals (0.5%, $n = 2$). Age distribution skewed toward younger professionals, with 277 (65.5%) participants <40 years of age, 125 (29.6%) aged 41-55 years, and 47 (11.1%) aged ≥56 years. Gender representation was balanced, with 48.8% ($n = 206$) male and 51.2% female ($n = 217$) respondents. Geographically, participants were distributed across Europe (45.4%, $n = 192$), Asia (27.0%, $n = 114$), the Americas (17.0%, $n = 72$), Africa (7.3%, $n = 31$), and Oceania (3.3%, $n = 14$), reflecting a global perspective on DH adoption in oncology.

Device availability and communication technologies

Most participants had access to workstations/laptops (89.1%, $n = 377$) and smartphones (69.0%, $n = 292$), while tablets were less common (25.5%, $n = 108$). For patient communication, email was most frequently used (51.8%, $n = 219$), followed by audio-only telehealth (42.3%, $n = 179$) and mobile chat services (40.4%, $n = 171$). Video telehealth was utilized by only 27.4% ($n = 116$), with 11.8% ($n = 50$) of respondents not using any digital communication tools.

Perceived effectiveness of technologies

Video telehealth received the highest effectiveness ratings, with 62.9% of users scoring it 4-5/5 for enabling complete communication tasks, compared with 50.3% for chat services and 42.3% for email. Audio telehealth showed moderate effectiveness (50.3% scoring ≥4), while SMS (32.8% ≥4) and institutional apps (51.9% ≥4) lagged (Table 2). EHRs and e-prescriptions were rated most effective among DH solutions, with 77.8% and 80.5% of users, respectively, scoring them 4-5/5. In contrast, rates of high effectiveness for telemonitoring (45.2% ≥4) and digital therapeutics (50.0% ≥4) were lower.

Training and adoption patterns

Only 28.2% ($n = 108$) of participants reported formal training in telemedicine systems, despite 46.8% ($n = 198$) of them reporting using them clinically. EHR training was more common (53.5%, $n = 205$), accompanied by higher adoption (51.3%, $n = 217$) and effectiveness ratings. E-prescriptions were also highly used (56.7%, $n = 240$) while other solutions were only scarcely used by <20% of participants. Most respondents (93.3%, $n = 237$) noted increased DH use over the last 24 months demonstrating that this is an actively and rapidly changing field, with 57.3% ($n = 203$) observing growing patient expectations. Adoption self-determination varied: 27.2% ($n = 93$) identified as active adopters, 32.7% as early supporters ($n = 112$), while 11.1% ($n = 38$) were late adopters or laggards.

Barriers and disparities

System integration challenges (67.4%, $n = 227$) and infrastructure gaps (57.3%, $n = 193$) were the foremost barriers. Older adults (82.5%, $n = 273$) and lower socioeconomic groups (70.1%, $n = 232$) were perceived as facing the greatest access hurdles. While 63.3% ($n = 215$) of participants felt confident (rated 4-5/5) in DH adoption, 40.7% ($n = 137$) cited inadequate training as a barrier. In response to education and practice resource needs of ESMO members on DH and AI, 84.1% ($n = 264$) of participants answered affirmatively demonstrating the potential utility of guidelines/recommendations on this topic.

Impact and future outlook

A majority (52.3%, $n = 171/327$) of participants agreed DH improved patient outcomes, with stronger consensus (80.2%, $n = 262/327$) on the fact that it will transform future care. Notably, 62.7% ($n = 205$) of participants strongly emphasized cybersecurity importance, reflecting growing awareness of data protection needs.

Table 2. Perceived effectiveness of different communication technologies, *n* (%)

	Not used (*n*)	Not effective at all 1	2	3	4	Highly effective 5
Audio-only telehealth	92	10 (3.8)	40 (15.2)	81 (30.7)	86 (32.6)	47 (17.8)
Video telehealth	170	5 (2.7)	17 (9.1)	47 (25.3)	76 (40.9)	41 (22.0)
Email	89	14 (5.2)	52 (19.5)	88 (33.0)	74 (27.7)	39 (14.6)
SMS	170	17 (9.1)	45 (24.2)	63 (33.9)	38 (20.4)	23 (12.4)
Chat services in mobile device	140	11 (5.1)	29 (13.4)	51 (23.6)	69 (31.9)	56 (25.9)
Institutional or commercial digital health app	227	9 (7.0)	20 (15.5)	33 (25.6)	43 (33.3)	24 (18.6)

Percentages based on users (excluding 'Not used').

Differences according to subgroups

The analysis revealed significant variations in DH adoption across demographic and professional environment, particularly according to age and geographic location. Clinicians >40 years of age showed substantially higher adoption of clinical-grade technologies, including a 12-percentage point advantage in video telehealth usage (35% versus 23%, $P = 0.014$). Furthermore, training disparities were detected, as 35% of older respondents referred to formal telemedicine instruction compared with just 24% of younger colleagues ($P = 0.028$), as in e-prescriptions (65% versus 52%, $P = 0.021$). Furthermore, younger doctors described in a higher degree data security (44% versus 31%, $P = 0.024$) and poor user experience and/or interface (41% versus 28%, $P = 0.019$) as barriers for incorporating digital solutions. There were no notable differences in responses or outcomes between male and female oncology professionals.

Geographic disparities revealed profound infrastructure-dependent adoption patterns. Participants from Europe, America, and Oceania enjoyed near-universal workstation access (93.8%, 90.3%, and 100% respectively), while their African counterparts faced significant resource constraints (67.7% access, $P < 0.001$). These material differences translated directly into tool preferences: high-bandwidth options like audio/video telehealth dominated in well-resourced regions (85.7% in Oceania, 72.4% in Europe), while text-based systems prevailed where connectivity was unreliable (63% chat-based consults in Africa) (Figure 1). E-prescriptions and EHR are scarcely used in Asia and Africa, where <50% reported having access to these technologies (Figure 2).

DISCUSSION

The integration of DH technologies into oncology practice reveals a transformation marked by both promise and paradox. While these tools offer unprecedented opportunities to enhance cancer care, their adoption remains uneven across regions and professional generations, exposing fundamental tensions between technological capabilities and health care realities. Our global survey uncovers a landscape where advanced solutions are embraced yet constrained by human, organizational, and systemic factors that vary across different contexts.

The preference for certain communication technologies reflects an evolving physician—patient relationship in the digital age. Synchronous communication modalities have not simply replaced in-person visits but rather created a new hybrid model of care that blends physical and virtual interactions.[18] This shift carries profound implications for the therapeutic alliance in oncology, where sensitive discussions about prognosis and treatment have traditionally relied on nonverbal cues and physical presence. While some studies suggest that telehealth can maintain or even enhance patient satisfaction,[19] particularly when clinicians adapt effectively to virtual platforms, our findings reveal an unexpected generational divide: older clinicians report higher adoption of these technologies than their younger counterparts, contrary to assumptions about digital natives' inherent adaptability. This paradox may reflect deeper generational differences in professional socialization, where older oncologists' experience in managing complex patient communication translates more effectively to virtual platforms, while younger clinicians may prioritize technical features over relational aspects.[20] Nevertheless, these data should be interpreted cautiously as health care professionals >40 years old are under-represented in the survey, and there could be a selection bias due to the method of survey dissemination in a subgroup of elderly professionals keener on technological solutions. In addition, the voluntary and anonymous nature of the survey introduces the possibility of self-selection bias, as professionals with greater interest or engagement in DH may have been more likely to respond.

The survey revealed a notable discrepancy between the perceived effectiveness of video telehealth and its actual utilization among oncology professionals. While 62.9% of respondents rated video telehealth as highly effective (scoring 4-5/5), only 27.4% reported using it in their practice. This gap may stem from several factors. Firstly, infrastructure limitations and system integration challenges, cited by 67.4% and 57.3% of respondents, respectively, could hinder widespread adoption, particularly in resource-constrained regions. Secondly, the lack of formal training in telemedicine systems (reported by only 28.2% of users) may contribute to lower confidence and practical barriers in implementing video telehealth. Additionally, workflow compatibility issues might explain the preference for simpler, asynchronous tools like email, which, despite lower effectiveness ratings, are more familiar and easier to integrate into existing routines.[21] The findings suggest that while video telehealth is recognized as a powerful tool for patient communication, its adoption is constrained by

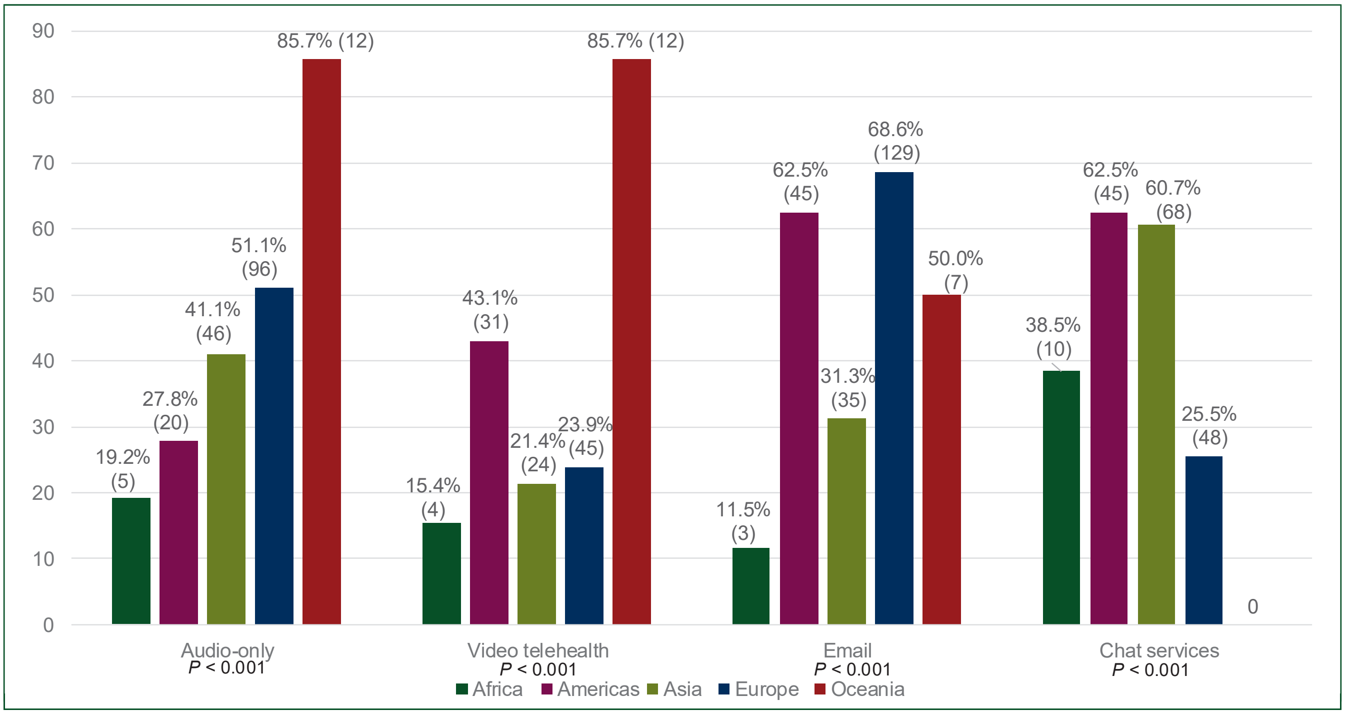

Figure 1. Regional patterns in digital communication tool utilization among oncology professionals. Bar chart showing the percentage of respondents using various communication technologies for patient interaction across five global regions (Africa, Americas, Asia, Europe, and Oceania). Technologies assessed include email, audio-only telehealth, chat/mobile messaging services, and video telehealth. Significant geographic variation was observed, with high-bandwidth tools (video telehealth) predominating in well-resourced regions, while text-based systems were more frequently used in regions with connectivity limitations. Global averages are represented by dashed horizontal lines for each technology category. Statistical comparisons revealed significant regional differences ($P < 0.001$ for all technology types, chi-square test).

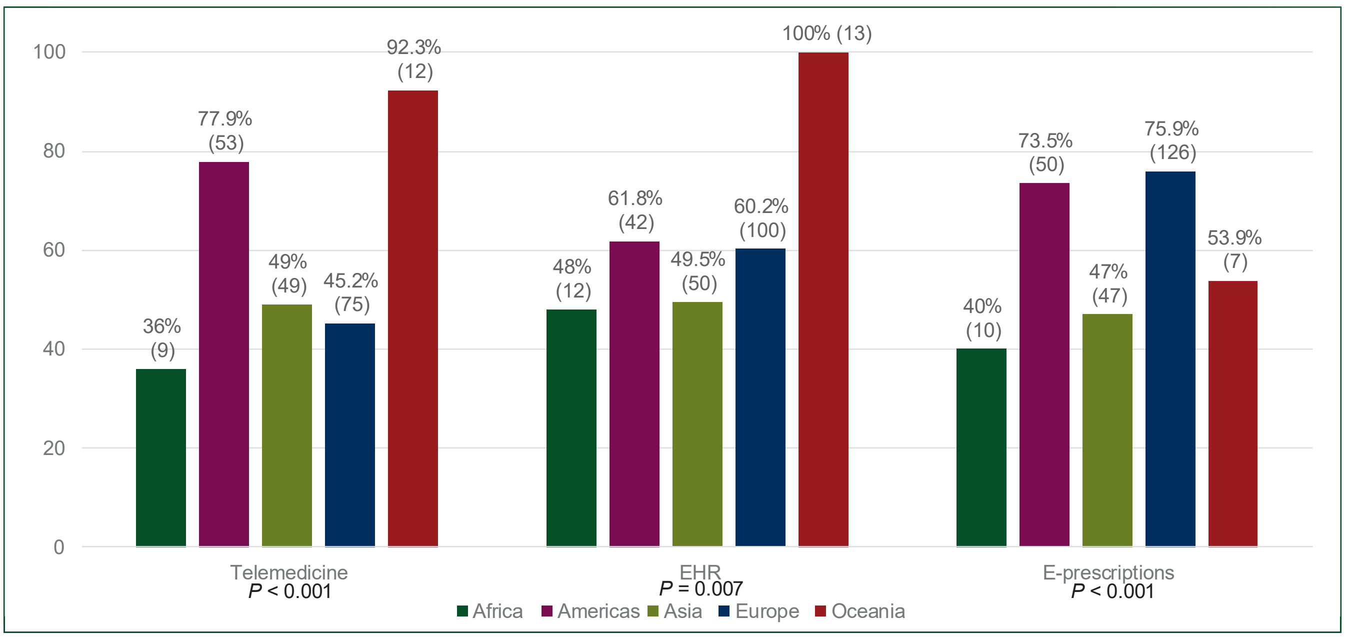

Figure 2. Geographic disparities in adoption of clinical digital health solutions. Bar chart illustrating the proportion of respondents reporting access to and utilization of key clinical digital health technologies stratified by continent. Solutions assessed include telemedicine, electronic health records (EHRs) and e-prescriptions. Marked regional inequities are evident. Data are presented as percentages of respondents within each region reporting active use of each digital solution. Statistical comparisons revealed significant regional differences (chi-square test).

systemic and educational barriers.[22,23] Addressing these challenges through targeted training, improved infrastructure, and better integration with clinical workflows could help bridge the gap between its potential and actual use.

The persistent use of email despite its perceived limitations reveals fundamental gaps in current DH ecosystems. Rather than representing mere inertia, this phenomenon may indicate unmet needs in asynchronous communication that newer platforms fail to address.[24] In the high-stakes environment of oncology care, in which information exchange often requires careful documentation and consultation, email provides a familiar, verifiable medium that fits existing workflows. This suggests that future DH solutions need to go beyond simply introducing novel functionalities, and they must genuinely understand and adapt to the nuanced communication patterns and medicolegal requirements of cancer care.[25] Besides, emails are free resources with no cost implication, a situation that could be involved in their wide application.

The identified barriers, particularly integration challenges and infrastructure limitations, expose critical flaws in how DH technologies are currently implemented. These are not simply technical problems, but manifestations of deeper issues in health care systems' capacity for innovation.[26] The fragmentation of digital tools across different platforms creates cognitive burdens for clinicians and risks to patient safety through information silos.[27] In resource-limited settings, these challenges are compounded by basic technological deficits, creating what might be termed 'digital determinism', in which the quality of cancer care becomes predetermined by the level of digital infrastructure available.[28] The striking continental disparities in workstation access reflect global inequalities that DH was supposed to mitigate but may instead be exacerbating through what has been called 'digital redlining', i.e. the systemic exclusion of certain populations from technological benefits.[29] Arora et al. underscored how such inequities are rooted in structural gaps, including broadband access and digital literacy, which disproportionately affect rural and minority populations.[30] For instance, Project ECHO's telementoring model demonstrates that even low-bandwidth solutions (requiring just 1.5 Mbps) can bridge specialist shortages in underserved areas, yet adoption remains limited without targeted policy interventions.[31] Similarly, the Accountability for Cancer Care through Undoing Racism and Equity (ACCURE) Pragmatic Quality Improvement Trial, a system-based intervention study addressing racial and ethnic inequities in early-stage breast and lung cancer treatment, highlighted how health informatics can mitigate bias and improve equity; however, its success hinges on integrating technology with human-centered supports like nurse navigators.[32] These examples reveal the paradox that while digital tools hold promise for democratizing care, their implementation often replicates existing hierarchies. Without deliberate efforts to address infrastructure gaps, training disparities, and systemic biases, DH may inadvertently entrench, rather than dismantle, the inequities it aims to resolve.

The identified educational needs point to a fundamental mismatch between technological advancement and professional development in oncology. Current medical education systems, largely designed for stable bodies of knowledge, struggle to keep pace with exponential technological change. This creates a competency gap where clinicians are expected to use tools they do not fully understand to make decisions with life-altering consequences.[33] The age-related differences in training exposure (older clinicians reporting more telemedicine training) suggest that current educational approaches may be misaligned with the needs of younger professionals who, despite greater digital literacy, face more complex technological ecosystems. Collaborative efforts between academic institutions, professional organizations, and industry stakeholders could yield standardized curricula that address the diverse needs of oncology professionals. ESMO AI & Digital Oncology Congress in 2025, held for the first time, is an example of this new resource need that could impact deeply in training and knowledge, specially designed specifically for oncologists.

The reported improvement in adverse event monitoring demonstrates the potential of DH to transform oncology practice beyond mere convenience.[8,9] By enabling continuous, data-rich patient monitoring, these technologies could shift cancer care from episodic to continuous, from reactive to proactive. However, this promise remains constrained by the 'last mile' problem of DH, i.e. the gap between technological capability and clinical implementation. Without addressing the human and organizational factors identified in our study (clinician confidence, workflow integration, equitable access), even the most sophisticated tools risk becoming expensive accessories rather than transformative solutions. Additionally, advancing data security measures and addressing privacy concerns will be critical for building trust in digital platforms.[34] Policymakers must ensure that regulatory frameworks keep pace with technological advancements, providing clear guidelines on data usage and protection.

These findings collectively suggest that the current approach to DH in oncology may be fundamentally flawed in its emphasis on technological innovation over organizational adaptation. True progress will require reconceptualizing DH not as a set of tools to be adopted, but as an ecosystem to be cultivated, one that balances technical capabilities with human factors, that prioritizes equity as much as innovation, and that recognizes the complex sociotechnical nature of health care transformation. This will demand nothing less than a new paradigm for DH implementation in oncology, where technology serves as a mean to more humane, equitable, and effective cancer care rather than an end in itself.

Conclusion

The present survey showed that DH technologies hold great promise for advancing oncology care, yet challenges in adoption and confidence remain. Tailored educational initiatives and improved infrastructure are vital to overcome these barriers and to ensure the widespread integration of DH solutions in oncology practice.

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DISCLOSURE

The authors have declared no conflicts of interest.

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